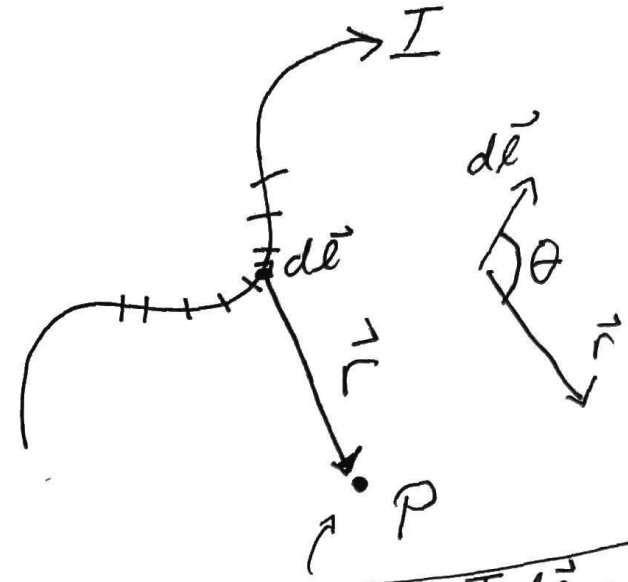
Biot-Savart Law

(Test)

(W: Faraday
M: Inductors
W: LR, LC Circuits)

Magnetic version of Coulomb's law. →



$$\boxed{d\vec{B} = \frac{\mu_0}{4\pi}\frac{I\,d\vec{\ell}\times\hat{r}}{r^2}}$$

- Point charge dQ → Infinitesimal current $d\vec{I}$

  $dQ = \lambda\, d\ell$, $\quad d\vec{I} = I\, d\vec{\ell}$

- Direction $\hat{r}$ → $d\vec{\ell}\times\hat{r}$

$$|dB| = \frac{\mu_0}{4\pi}\frac{I\,d\ell}{r^2}\sin\theta$$

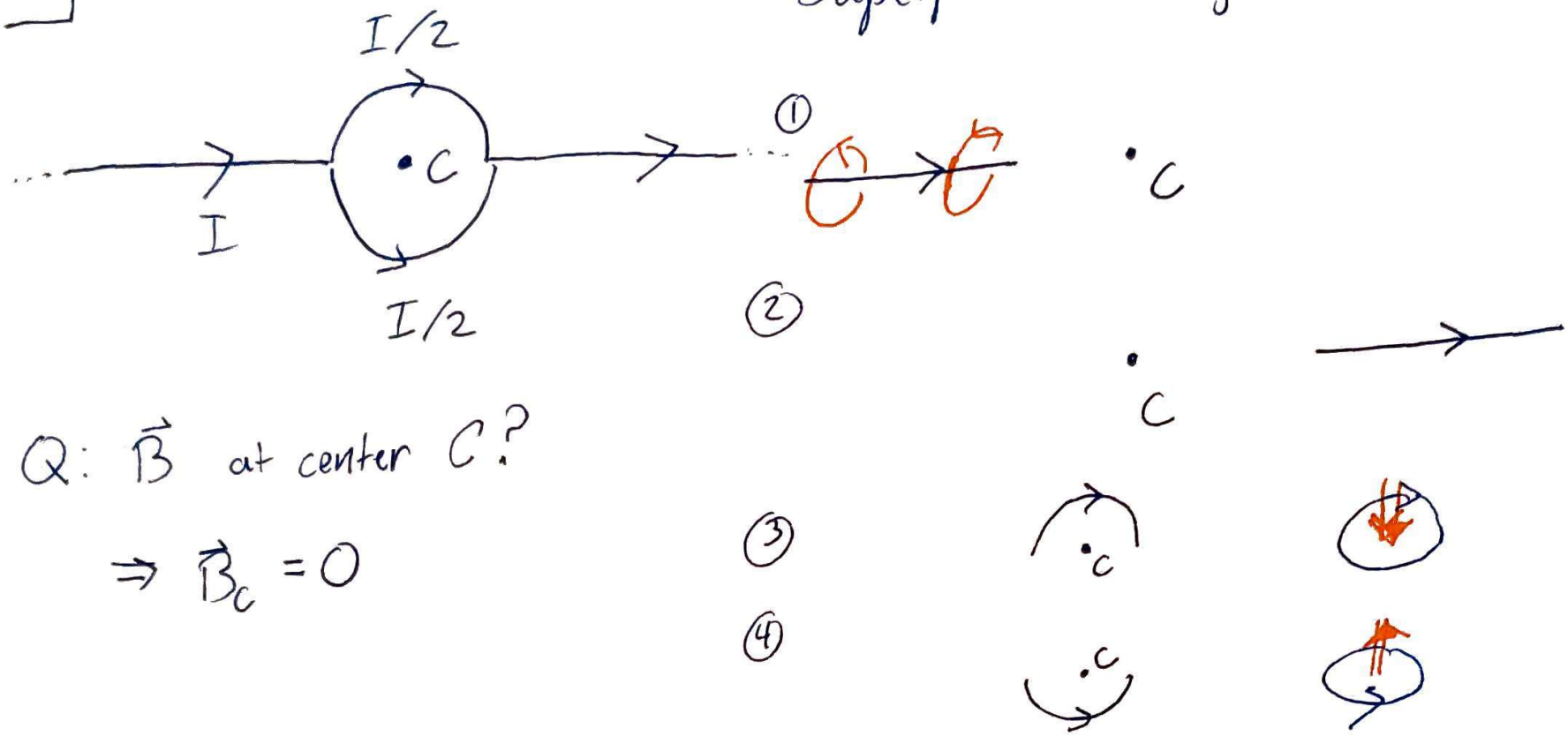
Perpendicular to $d\vec{\ell}, \vec{r}$ plane

↳ Often, just in or out of page.

Typical problem: Finite line of current, or circular current loops.

1

Superposition of:



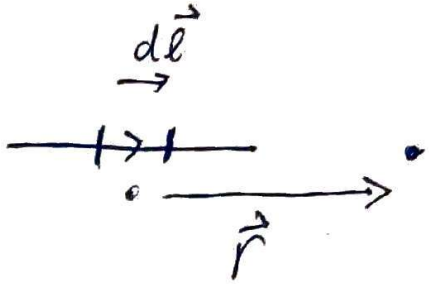
Q: $\vec{B}$ at center C?

$\Rightarrow \vec{B}_C = 0$

$\vec{B}$ from ①, ② each zero:

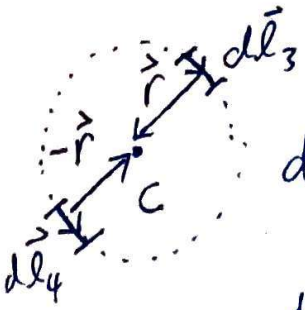
$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{\ell} \times \hat{r}}{r^2} = 0$$

$$d\vec{\ell} \times \hat{r} = 0$$



$(\sin\theta = \sin(0) = 0)$

$\vec{B}_{③} + \vec{B}_{④} = 0$

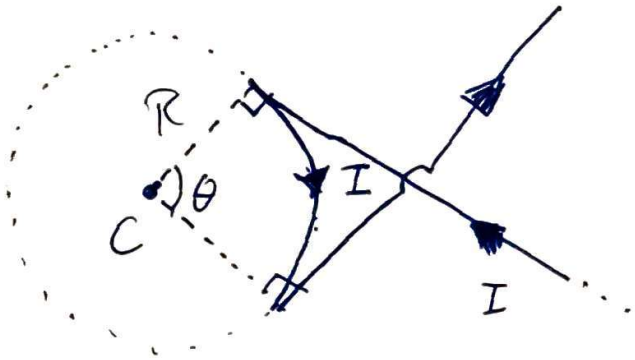


, $d\vec{\ell}_3 = d\vec{\ell}_4$

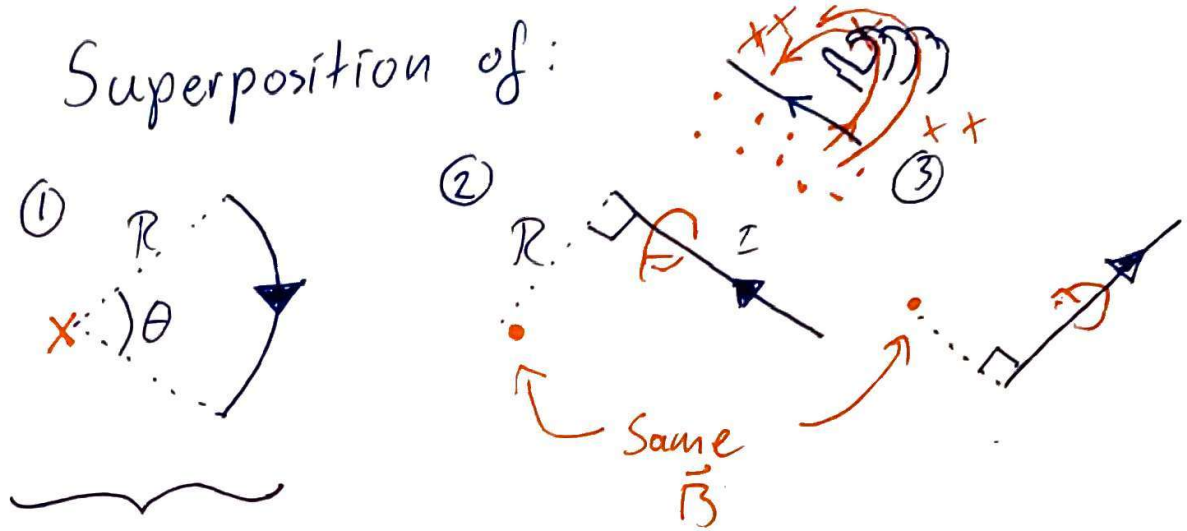
$$d\vec{B}_{net} = \frac{\mu_0 I}{4\pi} \frac{1}{r^2}\left(d\vec{\ell} \times \hat{r} + d\ell \times (-\hat{r})\right)$$

$$= \frac{\mu_0 I}{4\pi} \frac{1}{r^2} d\vec{\ell} \times (\hat{r} - \hat{r}) = 0$$

2)



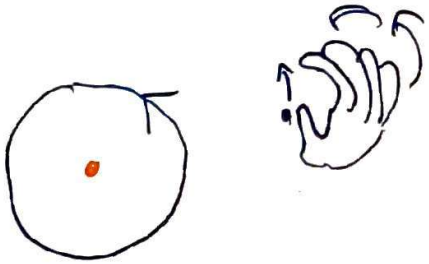
Superposition of:

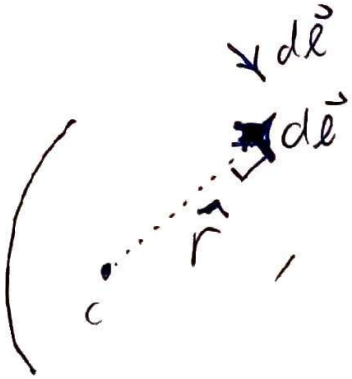


Q: For what θ is $\vec{B}_c = 0$?

For a complete circle: $\vec{B} = \dfrac{\mu_0 I}{2R}(-\hat{z})$

$\Rightarrow$ For the arc: $\vec{B} = \underbrace{\dfrac{\theta}{2\pi}}_{\text{Fraction of circle}} \vec{B}_{full} = \dfrac{\theta}{2\pi}\dfrac{\mu_0 I}{2R}(-\hat{z})$

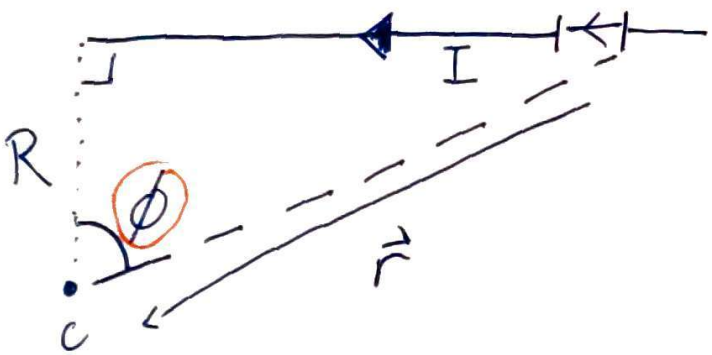




$d\vec{l} \times \hat{r} = -dl\,\hat{z}$

Everywhere on circle; each dl contributes same

- When applying Biot-Savart to lines/line segments, parameterize by an <u>angle</u>. (ϕ in this case)



$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l} \times \hat{r}}{r^2}$$

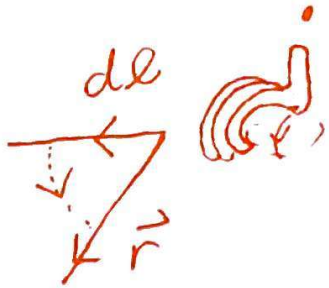
$$r = \frac{R}{\cos\phi}, \quad \phi \in [0, \tfrac{\pi}{2})$$

($\phi = \pi/2$, $\vec{r}$ is parallel to wire)

$$\frac{1}{r^2} = \frac{\cos^2\phi}{R^2} \checkmark$$

$$d\vec{l} \times \hat{r} = (\underset{|d\vec{l}|}{dl} \cdot \underset{|\hat{r}|}{1} \cdot \sin\theta)\,(+\hat{z})$$

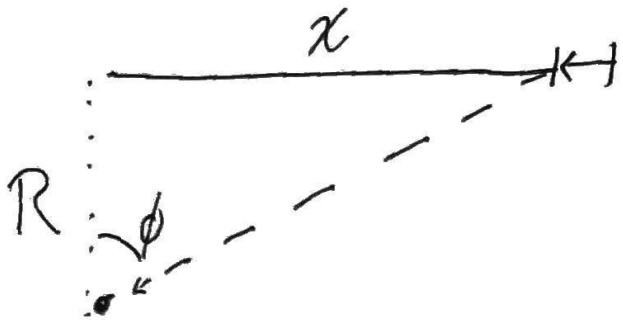
RHR



$$= \underbrace{dl}\ \cos\phi\ \hat{z}$$

↳ Rewrite in terms of ~~dl~~. $d\phi$.

$$\theta = (\pi - \pi/2 - \phi)$$
$$= \pi/2 - \phi$$



$$dl = dx$$

$$x = R\tan\phi$$

$$dl = dx = \frac{dx}{d\phi}\,d\phi = R\sec^2\phi\,d\phi = \frac{R}{\cos^2\phi}\,d\phi$$

$$\hookrightarrow d\vec{l}\times\hat{r} = dl\cos\phi\,\hat{z}$$

$$= \frac{R\,\hat{z}}{\cos\phi}\,d\phi \quad \checkmark$$

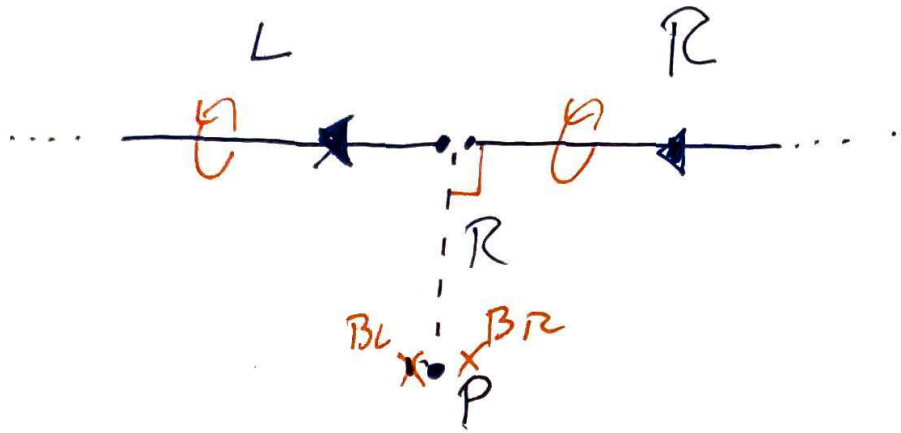
$$d\vec{B} = \frac{\mu_0}{4\pi} I \frac{d\vec{l}\times\hat{r}}{r^2} = \frac{\mu_0 I}{4\pi}\frac{R}{\cos\phi}\,d\phi\cdot\frac{\cos^2\phi}{R^2}\,\hat{z}$$

$$= \boxed{\frac{\mu_0 I}{4\pi}\frac{1}{R}\cos\phi\,d\phi\,\hat{z}}$$

$$\vec{B} = \int d\vec{B} = \int_0^{\pi/2} d\phi\,\frac{\mu_0 I}{4\pi}\frac{1}{R}\cos\phi\,\hat{z}$$

$$= \frac{\mu_0 I}{4\pi}\frac{1}{R}\hat{z}\underbrace{\int_0^{\pi/2}\cos\phi\,d\phi}_{1} = \boxed{\frac{\mu_0 I}{4\pi R}\hat{z}}$$

(Alternatively:



$$\vec{B}_P = \vec{B}_L + \vec{B}_R$$

$$= 2\vec{B}_R$$

Special to this setup.

But using Ampere's law:

$$\vec{B} = \frac{\mu_0 I}{2\pi R}\hat{z}$$

$$\Rightarrow \quad \vec{B}_R = \frac{1}{2}\vec{B} = \frac{\mu_0 I}{4\pi R}\hat{z}$$

$$\vec{B}_{①} = \frac{\theta}{2\pi}\frac{\mu_0 I}{2R}(-\hat{z}) = \frac{\mu_0 I}{4\pi R}\theta(-\hat{z})$$

$$\vec{B}_{②} = \frac{\mu_0 I}{4\pi R}\hat{z}$$

$$\vec{B}_{③} = \frac{\mu_0 I}{4\pi R}\hat{z}$$

$$\vec{B}_c = \vec{B}_{①} + \vec{B}_{②} + \vec{B}_{③}$$

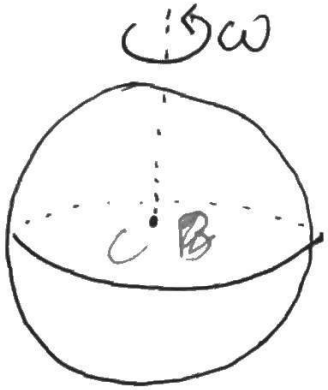
$$= \frac{\mu_0 I}{4\pi R}(1 + 1 - \theta)\hat{z}$$

$$= \boxed{\frac{\mu_0 I}{4\pi R}(2-\theta)\hat{z}} \Rightarrow B_c = 0$$

$$\Downarrow$$

$$\boxed{\theta = 2} \approx 114^\circ$$

3]



Spherical shell, rad R

Uniform charge σ

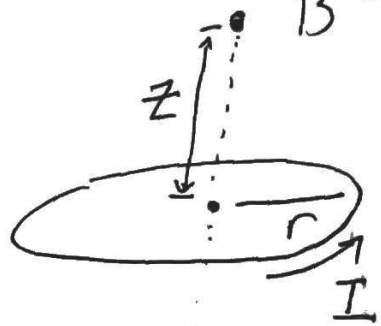
Spinning w/ ang. freq ω

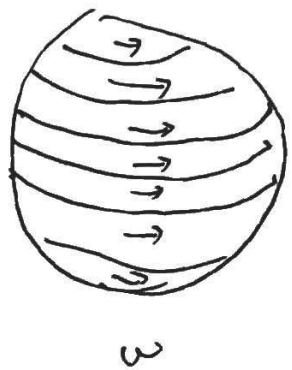
<u>Q</u>: $\vec{B}$ at center C?

Use:

$$\vec{B} = \frac{\mu_0 I r^2}{2(r^2+z^2)^{3/2}}\hat{z}$$

(B rot-savart)





Idea: Superposition of current loops.

<u>Caution</u>: <u>Sphere</u>

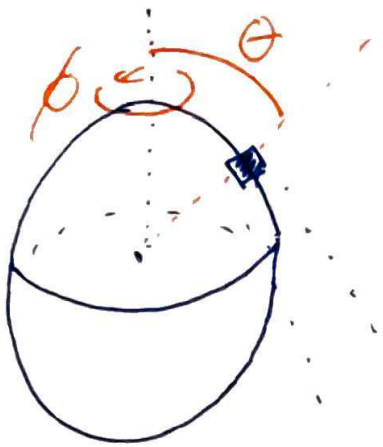
(not a cylinder)

Spherical coords: (θ, ϕ)

θ: Top to bottom

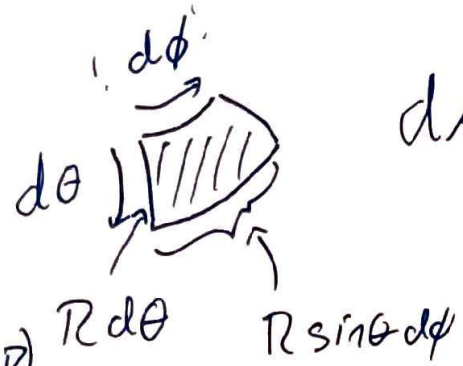
$\theta = 0 \qquad \theta = \pi$

ϕ: Around the axis $[0, 2\pi)$



Circle:

Careful about area element:

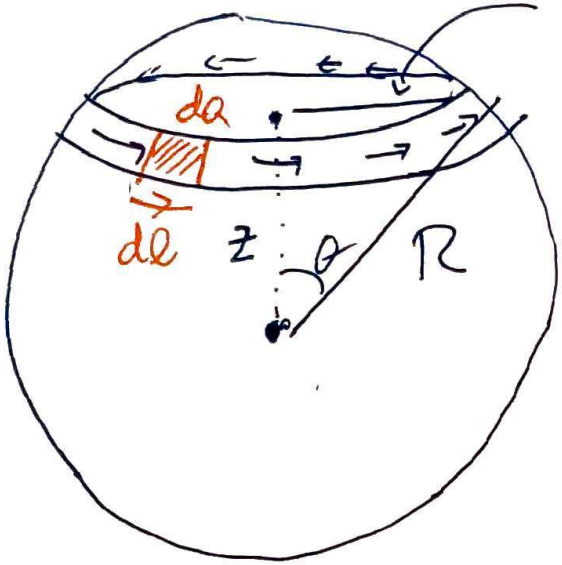


$$dA = R^2 \sin\theta \, d\theta \, d\phi$$

$$= (R\,d\theta)\cdot(R\sin\theta\,d\phi)$$

$$= (R\,d\theta)$$

$$r = \sqrt{R^2 - z^2} = R\sin\theta$$



Current loop:

r, z are good

$$d\vec{B} = \frac{\mu_0 I r^2}{2(\underbrace{r^2+z^2}_{R^2})^{3/2}}\hat{z}$$

$$= \frac{\mu_0 r^2}{2R^3}\hat{z}\ \underbrace{I}_{\text{Requires some care}}$$

Spherical coord

θ, ϕ

↳ How we'll add loops up to get a sphere.

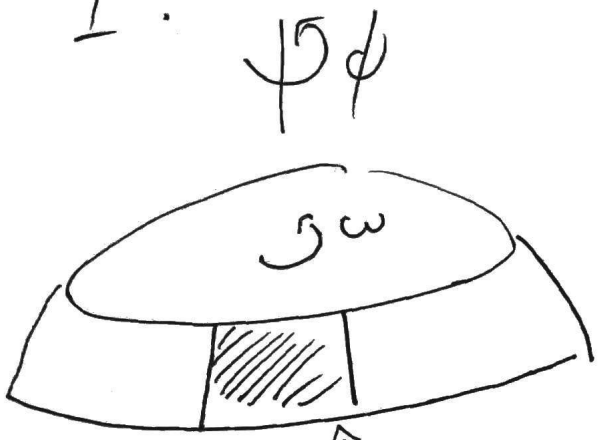
$$I = \lambda \cdot v = \underline{\underline{\lambda}}\, r\omega$$

Linear charge, $\dfrac{dQ}{dl} = \dfrac{(\sigma\, dA)}{dl}$ ~~$\sigma \sin\theta\, d\theta$~~

$$= \frac{\sigma\ R^2 d\theta\ \sin\theta\, d\phi}{R\sin\theta\, d\phi} = \sigma R\, d\theta$$

$$\boxed{\begin{aligned} I &= (\sigma R\, d\theta)\, r\omega \\ &= \sigma R^2 \omega \sin\theta\, d\theta \end{aligned}}$$

I ?



$$I = \cancel{dq} \cdot v = \cancel{dq}$$

$$\lambda \qquad \lambda \, \omega r$$

$$dq = \sigma \, dA$$

$$I = \lambda \, dl \leftarrow \quad dl = r \, d\phi = R \sin\theta \, d\phi$$

$\vec{dl}$

$$dA = R^2 \sin\theta \, d\theta \, d\phi$$

$$\sigma \, dA = \lambda \, dl$$

$d\theta$

$$\sigma \, \underline{R^2 \sin\theta \, d\theta \, d\phi} = \lambda \, \underline{R \sin\theta \, d\phi}$$

I flowing past segment $d\theta$

$$\boxed{\sigma R \, d\theta = \lambda}$$

$$dB = \frac{\mu_0 I r^2}{2R^3}\hat{z}, \qquad I = \sigma R^2 \omega \sin\theta \, d\theta$$

$$r^2 = R^2 \sin^2\theta$$

$$\Rightarrow \quad d\vec{B} = \frac{\mu_0 \sigma \omega R}{2} \sin^3\theta \, d\theta \, \hat{z}$$

$$\sin^2 = 1 - \cos^2$$

$$\Rightarrow \vec{B} = \int d\vec{B} = \int_0^{\pi} \mu_0 \frac{\sigma \omega R}{2} \sin^3\theta \, d\theta \, \hat{z}$$

$$\sin^3 = \sin - \cos^2 \sin$$

$$= \left(\mu_0 \frac{\sigma \omega R}{2} \hat{z}\right) \underbrace{\int_0^{\pi} \sin^3\theta \, d\theta}$$

$$\int_0^{\pi} \sin\theta \, d\theta - \int_0^{\pi} \sin\theta \cos^2\theta \, d\theta$$

$$u = \cos\theta \rightarrow du = -\sin\theta \, d\theta$$

$$2 - \int u^2 du = 2 + \left[\frac{\cos^3\theta}{3}\right]_0^{\pi} = 2 - \frac{2}{3}$$

$$= 4/3$$

$$\boxed{\vec{B} = \frac{2}{3}\mu_0 \sigma \omega R \hat{z}}$$